

PRINCE OMUYEH

484-318-5778 | princeomuyeh@gmail.com | linkedin.com/in/prince-omuyeh | peo226.github.io

EDUCATION

Villanova University

Master of Science in Software Engineering

Villanova, PA

August 2026 – Present

Lehigh University, P.C. Rossin College of Engineering

Bachelor of Science in Computer Science, Minor in Data Science

Bethlehem, PA

August 2022 - May 2026

Relevant Coursework: AI Theory & Practice, Statistical Computing, Data Mining, Database Systems, Distributed Systems, Data Structures & Algorithms, Operating Systems Design

SKILLS

Languages: Python, R, SQL, Java, C++, JavaScript, Rust, HTML/CSS

Frameworks & Libraries: TensorFlow, Keras, scikit-learn, Pandas, Ultralytics, React, Spring, Maven, Node.js

Developer Tools: Git, Docker, Linux, TensorBoard, Kafka, PostgreSQL, Jira/Atlassian, Agile/Scrum

EXPERIENCE

Computer Vision Engineer

January 2025 – December 2025

Hawkeye (Capstone Sponsor)

Bethlehem, PA

- Improved a real-time Python computer vision pipeline for cyclist hazard detection, collaborating with a startup sponsor on a 4-person Agile team via Jira
- Migrated the object detection model from YOLOv8 to YOLOv11 by updating model architecture configs, refactoring preprocessing and postprocessing logic, and resolving dependency changes across the Ultralytics codebase
- Achieved a vehicle orientation prediction accuracy of 92.66%

Software Development Intern

March 2025 – May 2025

Lehigh University

Bethlehem, PA

- Built an admin interface for a full-stack academic integrity platform using JavaScript and React, enabling 100+ faculty and associate chairs to manage user roles and control access to student reports
- Implemented 6 RESTful APIs and OAuth authentication supporting role-based access control for faculty, associate chairs, and student accounts

PROJECTS

Distributed Replicated Hash Table | C++

April 2026

- Implemented a distributed hash table across 5 nodes with Paxos-based consensus, supporting atomic operations, data replication, and fault tolerance
- Achieved ~30,000 ops/sec throughput and sub-1ms average latency at 10k keys under fault injection testing, maintaining availability after node crashes with no manual intervention

Classification Algorithms Analysis | Python, scikit-learn, SVM

April 2025

- Implemented and evaluated SVM classification across MNIST, CIFAR-10, and Iyer datasets using scikit-learn, performing hyperparameter tuning with GridSearchCV (C and γ)
- Evaluated model performance via accuracy, F1 score, and AUC metrics; conducted sensitivity analysis to identify optimal configurations per dataset

JPMorgan Chase Transaction Processing System | Java, Spring, Kafka, SQL

April 2025

- Designed a Kafka-based transaction processing system in Spring Boot backed by an H2 SQL database
- Developed transaction pipelines, external API interactions, and user balance endpoints using Java, Maven, and Spring

Financial Forecasting Model | R, Linear Regression, Supervised Learning

May 2024

- Built a predictive model for the U.S. 10-year Treasury yield using linear regression and supervised learning, leveraging historical macroeconomic data (inflation, GDP, unemployment)
- Applied feature selection and data cleansing techniques across multiple economic indicators, achieving a 15% improvement in prediction accuracy over baseline models